Decision Log

LendingClub loan default project | DATA 230 group project | Last updated October 9, 2026

> A running record of the decisions made on this project. Each entry states the problem, the options that were weighed, the option chosen, and the reason. Numbers come from the notebook lending_club_1st_draft.ipynb, run on the full file (2,260,701 rows, 151 columns).

Decisions made

D1 What the project predicts

Stage: project direction | Decided October 8, 2026

PROBLEM

The dataset supports several machine learning tasks, and the project needs one clear target before the EDA starts. The group's first idea was to predict the approved loan amount.

OPTIONS

1. Regression on loan amount.
2. Regression on interest rate.
3. Clustering borrowers into segments.
4. Classification: Charged Off versus Fully Paid.

CHOSEN

Option 4.

Binary classification of whether a loan is charged off or fully paid, using only information known at application.

REASON

- It is a real outcome: 1,076,751 loans were fully paid and 268,559 were charged off.
- The 20% charge-off rate is a class imbalance, which is the assignment brief's own example of an EDA finding that justifies an ML direction.
- Loan amount is a weak target: it is what the borrower requested, the file holds only accepted loans, and the amount is capped at $40,000.
- Interest rate is nearly determined by the lender's grade, and clustering has no ground truth to evaluate against.

D2 Where the analysis runs

Stage: environment | Decided October 9, 2026

PROBLEM	The data file is 1.6 GB. The grading rubric mentions GPU-accelerated EDA (RAPIDS), but the Google Colab GPU runtime restarted three times while running the first inspection step.
OPTIONS	1. Google Colab with a T4 GPU and cudf.pandas. 2. Local Jupyter through Anaconda, on the CPU.
CHOSEN	**Option 2, for now.** All work runs locally on the CPU. GPU code will be written later if required.
REASON	• The local notebook completed every Stage 1 check without crashing; the Colab session did not. • Open point: the rubric line on GPU-accelerated EDA is not yet addressed and will need a decision before submission.

D3 Income and debt-to-income for joint applications

Stage 1, Step 2 (what is broken) | Decided October 9, 2026

PROBLEM

1,667 loans report an annual income of zero, and 1,659 of those have no dti. Another 2,561 loans have a dti above 100. Every one of these loans is a joint application, and the two groups do not overlap. The individual applicant's figures are not the ones a joint loan was judged on.

OPTIONS

1. For joint applications, use the joint figures (annual_inc_joint, dti_joint) in place of the individual ones, and keep application_type as a feature.
2. Keep the individual figures for every loan.
3. Drop joint applications from the analysis.

CHOSEN

Option 1.

Build one combined income column and one combined dti column that take the joint value for joint applications and the individual value otherwise. Keep application_type so the model can tell the two kinds of loan apart.

REASON

- On a joint loan both borrowers repay, so the joint figures measure the income and debt that back the loan.
- dti is a ratio, so a joint value means the same thing as an individual one.
- Option 2 leaves zero incomes and dti values above 100 that are meaningless; option 3 loses a whole segment (5.3% of all loans).

Known limitations

- Combined income is not comparable per person: the joint column averages about $123,600 against about $78,000 for the individual column. The new column means "income backing the loan".
- The meaning of the joint columns should be confirmed against LendingClub's data dictionary.
- To be validated with a chart of charge-off rate against dti, split by application type.
- 135 loans with a dti of 999 are not explained by this and are still treated as a placeholder value.

Consequence

annual_inc_joint and dti_joint are about 95% empty but must not be dropped until the combined columns are built.

D4 Which loans count as having an outcome

Stage 1, Step 4 (define the target) | Decided October 9, 2026

PROBLEM

The file has nine loan statuses, and only some of them are final. The target needs loans whose outcome is known: 878,317 loans are still Current, and 34,252 are late or in a grace period.

OPTIONS

1. Keep only Fully Paid and Charged Off; drop every other status and the 33 junk rows.
2. Also count Late (31-120 days) and Default as bad loans.
3. Also fold the "does not meet the credit policy" loans into the two classes.

CHOSEN

Option 1.

Keep Fully Paid (class 0) and Charged Off (class 1) in a new data frame, with a 0/1 target column named charged_off. The raw data frame is left untouched.

REASON

- Only these two statuses are a known final outcome. A late loan can still recover or be charged off, and a target cannot be imputed.
- The "does not meet the credit policy" loans (2,749, about 0.2%) were approved under older rules, so they are a different population.
- The result is 1,345,310 loans (1,076,751 fully paid and 268,559 charged off, a 20.0% charge-off rate); 915,391 rows are dropped.

Known limitations

- Dropping Current loans removes 39% of the file and skews recent years toward loans that ended early. The year-by-year table in Step 4 measures this.
- Late (31-120 days) loans, 21,467 of them, mostly end as charge-offs; leaving them out is the cautious choice, not the only defensible one.

D5 Keep only loans old enough to have finished

Stage 1, Step 4 (define the target) | Decided October 9, 2026

PROBLEM

The data was captured in March 2019, so recent loans have not had time to finish. A recent loan that is already Fully Paid or Charged Off ended early, which makes the finished loans of recent years an unfair sample. For 60-month loans the charge-off rate among finished loans is 23.6% to 27.7% in the complete years but 36.4% for 2015 and 2016.

OPTIONS

1. Keep all years and state the bias as a limitation.
2. Remove every loan issued in the last 36 months before the capture date.
3. Keep a loan only if its own full term (36 or 60 months) had passed by the capture date.

CHOSEN

Option 3.

36-month loans are kept if issued through March 2016, and 60-month loans if issued through March 2014.

REASON

- It is one rule that handles both terms. A single 36-month cut would leave 60-month loans from April 2014 to March 2016 in the data while still incomplete.
- Of all loans in the kept age range, 98.8% are finished, so the outcomes are no longer selected by ending early.
- The result is 776,276 loans (658,666 fully paid and 117,610 charged off); 569,034 finished loans are removed.

Consequences

- The charge-off rate becomes 15.2%, not 20.0%. The 20% figure was inflated by the incomplete years, so the class imbalance quoted for the ML direction changes.
- 60-month loans drop to 67,908 (8.7% of the data), all issued from 2010 to March 2014.
- The most recent loans are lost, so the model is built on older lending. Loans from the first quarter of 2016 sit at the cutoff and may carry a small leftover of the bias.

D6 Free-text and location columns

Stage 1, Step 5 (clean the data) | Decided October 9, 2026

PROBLEM

Four text columns have many distinct values: emp_title (272,023 on the 776,276 loans kept), title (61,678), zip_code (924) and addr_state (51). A column with that many values cannot be used as a category without work, but some of them carry real information.

OPTIONS

1. Drop emp_title, title and zip_code; keep addr_state.
2. Drop emp_title and title; keep zip_code and addr_state.
3. Keep all four and clean the free text.

CHOSEN

Option 2.

Drop emp_title and title. Keep zip_code and addr_state.

REASON

- emp_title is hand-typed: 224,000 of its 272,023 values appear exactly once, and the same job has many spellings (Registered Nurse, RN, rn).
- title is a near-copy of purpose: the most common title for each purpose is that purpose in words, and the rest is hand-typed.
- zip_code is a three-digit ZIP area with real signal: among the 402 areas with at least 500 loans, the charge-off rate runs from 8.7% to 21.3%, against 15.2% overall.
- addr_state is clean and complete, and Tableau can map it directly.

Still open

- Whether zip_code goes on a Tableau map or into the model is left for later, if time allows. A three-digit ZIP map needs an outside lookup file; 168 of the 924 areas have fewer than 100 loans.

D7 Columns that are more than 30% empty

Stage 1, Step 5 (clean the data) | Decided October 9, 2026

PROBLEM

On the 776,276 loans kept, 58 columns are more than 30% empty. Most are empty for a structural reason: the field applies only to some loans (hardship plans, settlements, joint applications), was not collected in the early years, or is blank because an event never happened.

OPTIONS

1. Drop all 58 columns.
2. Drop them, but turn the five "months since last delinquency or record" columns into yes/no flags first.
3. Keep the structurally empty columns and impute them.

CHOSEN

Option 1.

Drop all 58, including the five "months since last delinquency or record" columns. The combined income and dti columns (decision D3) are built from the joint columns before those are dropped.

REASON

- It follows the lecture's rule for columns over 30% missing: too little information to rely on.
- The emptiest groups are 98% to 100% empty on these loans: joint and second-applicant fields, hardship fields, settlement fields, next payment date and member_id.
- The 14 newer credit-bureau fields are 86.5% to 88.4% empty, because the loans kept are mostly older than those fields.
- The "months since" columns are 51.3% to 84.2% empty, and columns such as delinq_2yrs and pub_rec already record whether those events happened.

Known limitation

- A blank "months since" value means the event never happened, which is information. Dropping the columns gives that up; option 2 would have kept it.

D8 Leakage columns

Stage 1, Step 5 (clean the data) | Decided October 9, 2026

PROBLEM

Seventeen columns describe what happened after the loan was issued: payments received, money recovered, the latest FICO score, last payment dates, and flags for payment plans, hardship and settlement. They would not be known when deciding on an application, and they give the outcome away.

OPTIONS

1. Drop all of them.
2. Keep them as features.

CHOSEN

Option 1.

Drop all 17 leakage columns.

REASON

- The goal is to predict the outcome from what is known at application, so anything recorded later cannot be a feature.
- Money was recovered after charge-off on 78.7% of charged-off loans and 0.0% of fully paid loans.
- Fully paid loans repaid 99.9% of the loan amount on average; charged-off loans repaid 39.8%.
- The latest FICO score averages 700.9 for fully paid loans and 573.9 for charged-off loans. At application the two groups were 701.1 and 690.7, so the gap opened after the loan was issued.

D9 How to fill missing values in numeric columns

Stage 1, Step 5 (clean the data) | Decided October 9, 2026

PROBLEM

After the drops, 39 numeric columns still have missing values on the 776,276 loans kept. Thirty-four are detailed credit-bureau fields at 6.1% to 16.1% missing, almost entirely because those fields were not collected before March to August 2012. Five more columns are under 0.1% missing.

OPTIONS

1. Fill skewed columns with the median and roughly symmetric columns with the mean.
2. Fill every column with the median.
3. Remove the 68,355 loans issued before September 2012, so no era has to be filled.
4. Drop the 34 credit-bureau fields, or leave the values blank.

CHOSEN

Option 1.

A column is treated as roughly symmetric when its skew is between -0.5 and 0.5, and is filled with its mean; every other column is filled with its median. A yes/no column, bureau_not_collected, marks the 67,527 loans whose bureau fields were never collected.

REASON

- In a skewed column the mean is pulled toward the long tail. For tot_cur_bal the mean is $132,392 and the median $69,424, so the median is the more typical value.
- In a symmetric column the mean and median are nearly equal, and the mean uses every value. Four columns qualify: mo_sin_old_il_acct, bc_util, percent_bc_gt_75 and revol_util. The other 35 are filled with the median.
- All of these columns are under 30% missing, the lecture's band for imputing, and filling keeps all 776,276 loans. Removing the early loans would have shrunk the data again.
- The early loans behave like the rest: they charged off at 15.3% against 15.1% for later loans, so filling them with a typical value does not create a false pattern.

Known limitations

- The gap is one era, not random. The early loans all receive the same value in about 30 fields, so those fields carry no real information for them. Charts of these fields should leave out the marked loans.
- mths_since_recent_inq is filled with its median of 5 months, but a blank there means no recent inquiry, and those borrowers charge off less (10.5% against 15.7%). The fill understates that.
- For the model, the fill values should be recalculated on the training data only.
- emp_length is text and is not filled by this rule; it is still open.

Proposed, not yet confirmed

ISSUE	PROPOSED DECISION	REASON
Lender's rating columns: grade, sub_grade, int_rate, installment	Exclude from the model's features; keep in the data for the EDA and use charge-off rate by grade as the benchmark.	They encode the lender's own risk verdict, so a model using them would largely repeat it.
home_ownership values ANY, OTHER, NONE	Merge into a single OTHER category.	They come from different periods of the application form and total only 478 finished loans.
Remaining rows of the Step 2 problem list	Apply the proposed fixes listed in the notebook: drop identifier and single-value columns, convert text to numbers and dates, treat placeholder values as missing.	Each fix follows from a check in Stage 1, Step 2.

These move to "Decisions made" once the group confirms them.